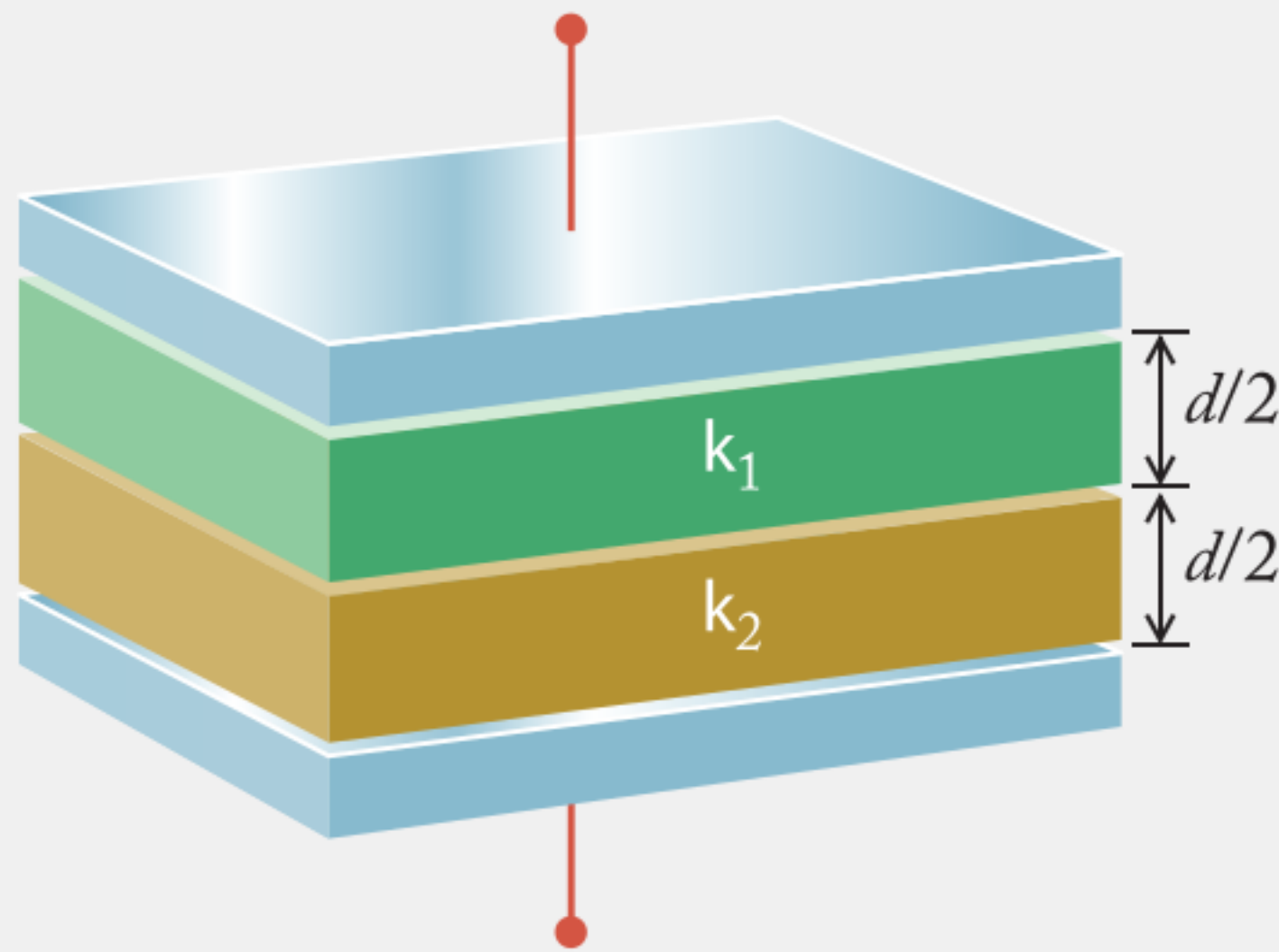


- *36. A parallel-plate capacitor of plate area A and spacing d is filled with two parallel slabs of dielectric of equal thickness with dielectric constants κ_1 and κ_2 , respectively (Fig. 26.33). What is the capacitance? (Hint: Check that the configuration of Fig. 26.33 is equivalent to two capacitors in series.)



- *37. A capacitor with two large parallel plates of area A separated by a distance d is filled with two equal-size slabs of dielectric side by side (Fig. 26.34). The dielectric constants are κ_1 and κ_2 . What is the capacitance?

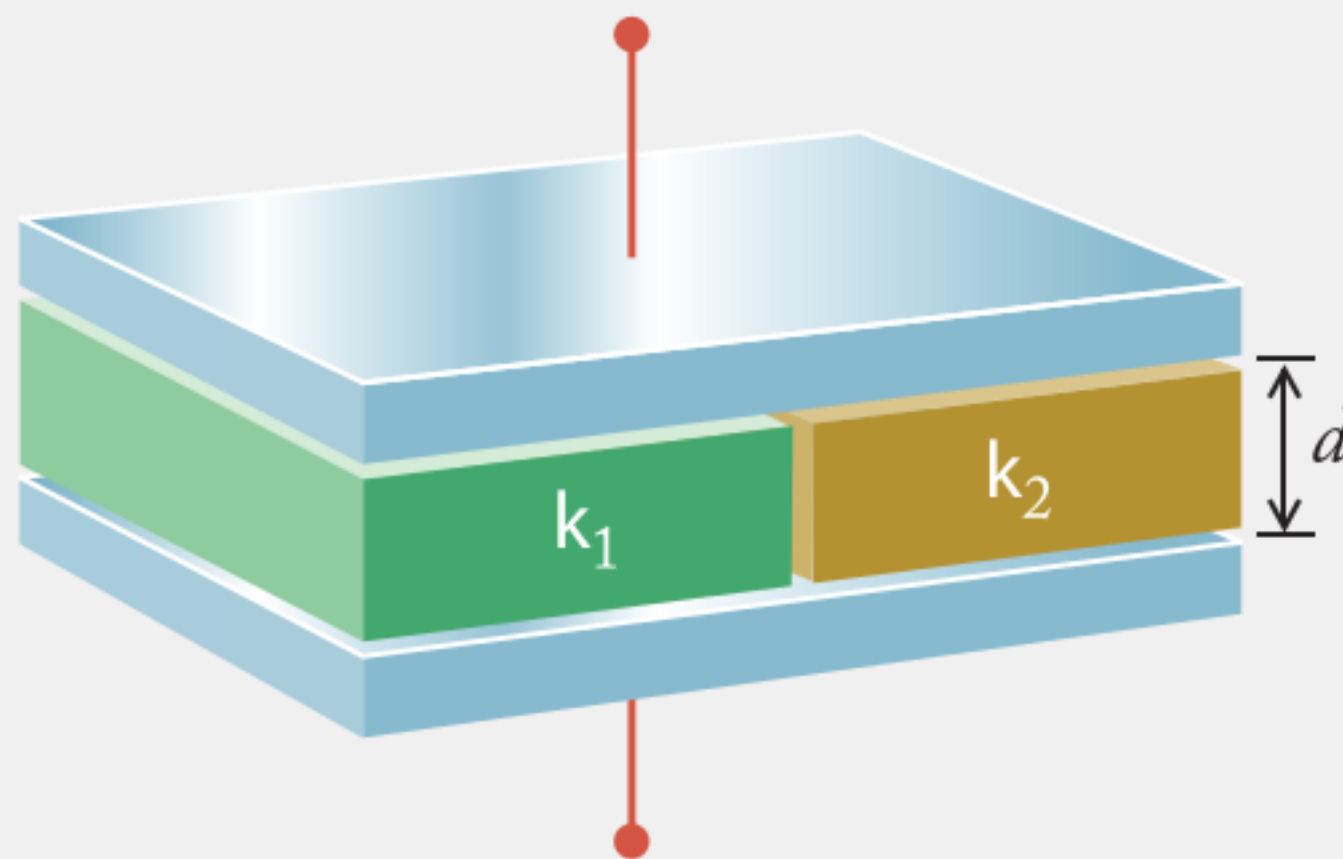


FIGURE 26.34 Parallel-plate capacitor with two slabs of dielectric side by side.

- 26.** Find (a) the equivalent capacitance of the capacitors in Figure P26.25, (b) the charge on each capacitor, and (c) the potential difference across each capacitor.

Figure P26.25

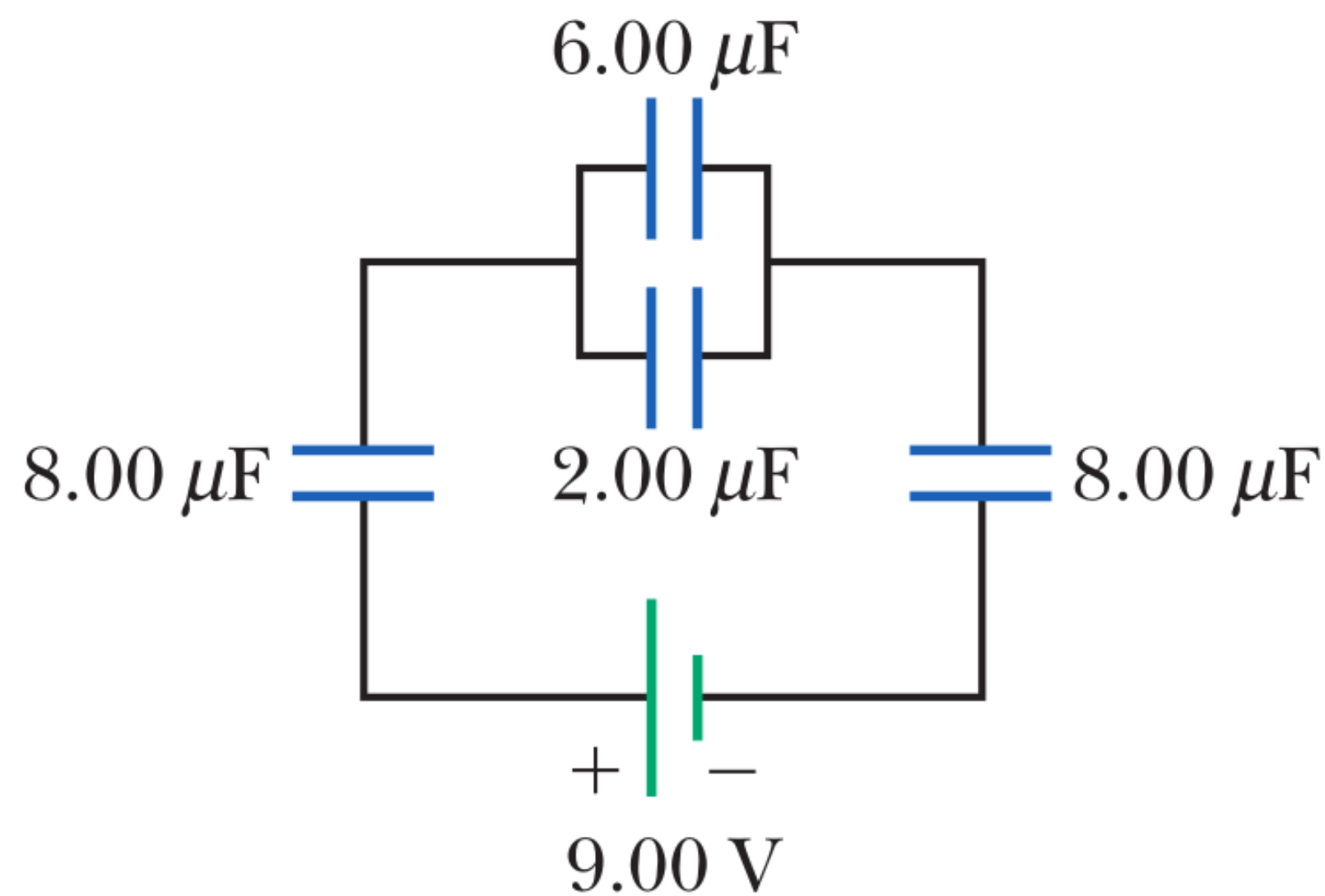


Figure P26.26

- *63. Three capacitors are connected as shown in Fig. 26.41. Their capacitances are $C_1 = 2.0 \mu\text{F}$, $C_2 = 6.0 \mu\text{F}$, and $C_3 = 8.0 \mu\text{F}$. If a voltage of 200 V is applied to the two free terminals, what will be the charge on each capacitor? What will be the energy in each?

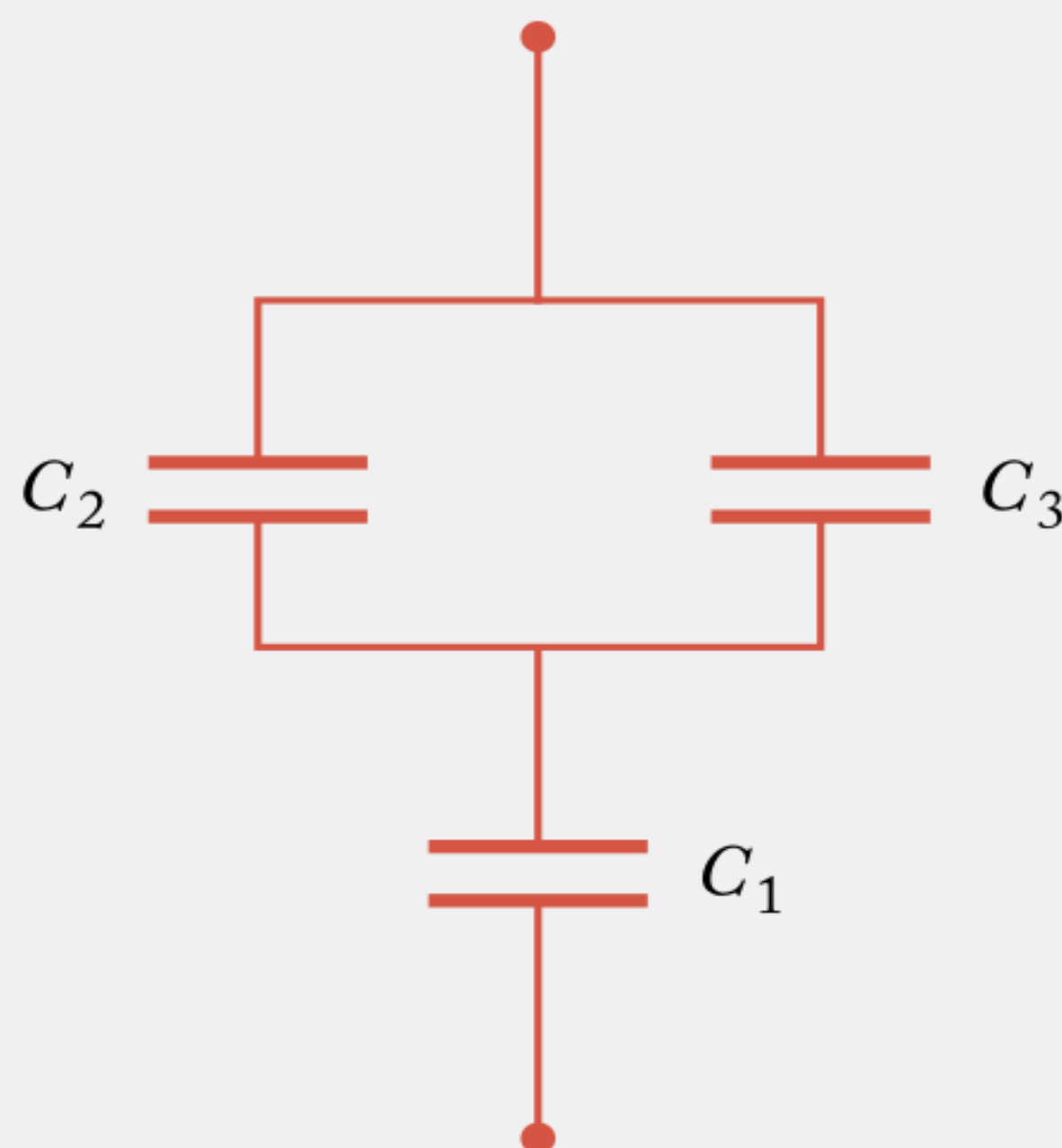


FIGURE 26.41 Three capacitors.

- 58.** Two large, parallel metal plates, each of area A , are oriented horizontally and separated by a distance $3d$. A grounded conducting wire joins them, and initially each plate carries no charge. Now a third identical plate carrying charge Q is inserted between the two plates, parallel to them and located a distance d from the upper plate as shown in Figure P26.58. (a) What induced charge appears on each of the two original plates? (b) What potential difference appears between the middle plate and each of the other plates?

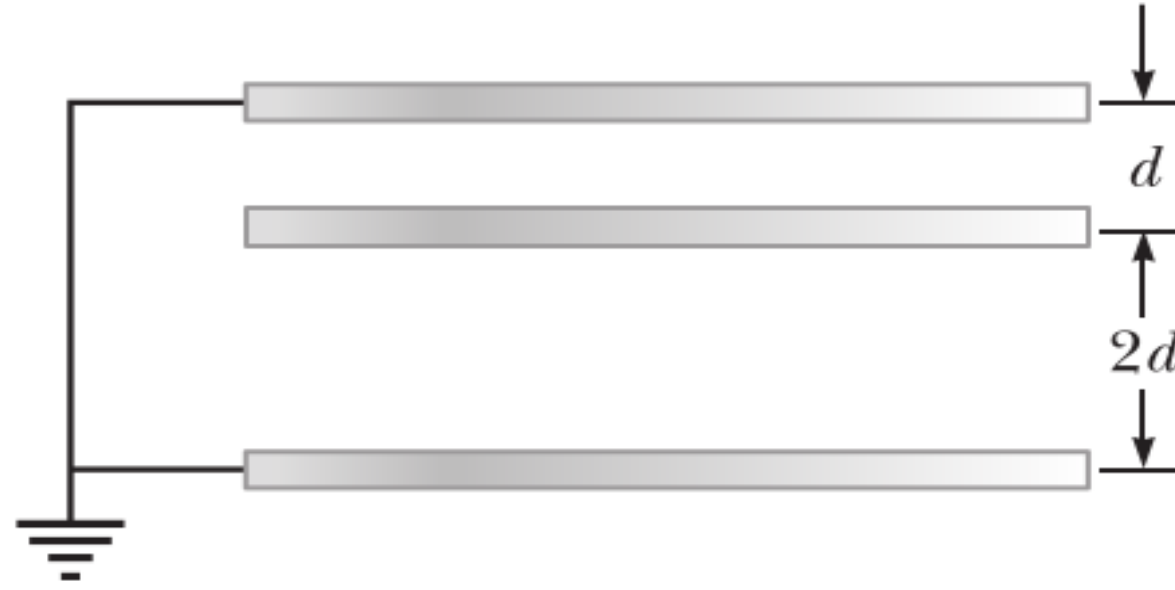


Figure P26.58

- *44.** A spherical capacitor consists of a metallic sphere of radius R_1 surrounded by a concentric thin metallic spherical shell of radius R_2 . The space between R_1 and R_2 is filled with a dielectric having a constant κ . Suppose that the free surface-charge density on the metallic sphere at R_1 is $\sigma_{\text{free}(1)}$.
- (a) What is the free surface-charge density on the metallic shell at R_2 ?
 - (b) What is the bound surface-charge density on the dielectric at R_1 ?
 - (c) What is the bound surface-charge density on the dielectric at R_2 ?